AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2018

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

2018 ing Only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

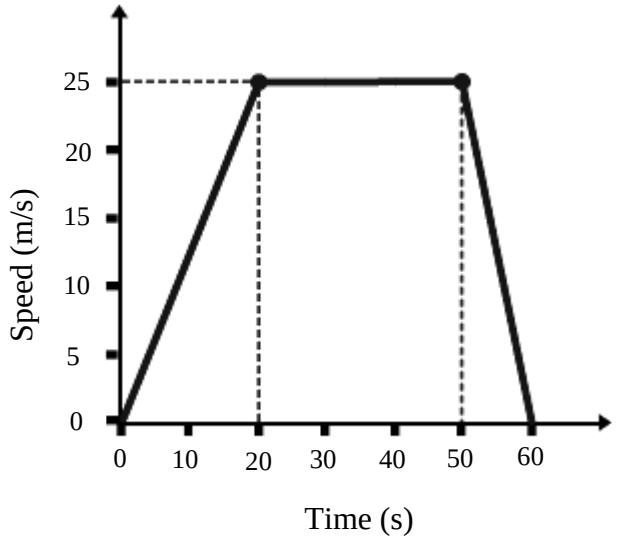
5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

'Physics helps in the development of our society.' In light of the given statement, describe any THREE important roles of physics for the development of society.

Q.2. (Total 3 Marks)

The given speed-time graph shows the journey of a motorcyclist.



a. Calculate the acceleration of the motorcyclist in the first 20 seconds. (2 Marks)

b. The motorcyclist has constant speed for how many seconds? (1 Mark)

Q.3. (Total 3 Marks)

A car has mass of 1000 kg and it moves in a circular path of radius 50 m with a constant speed of 25 m/s. Calculate the centripetal force of the car.

Q.4. (Total 3 Marks)

The centre of gravity of a hollow sphere always lies within the hollow region of the object. Justify the given statement.

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Q.5. (Total 3 Marks)

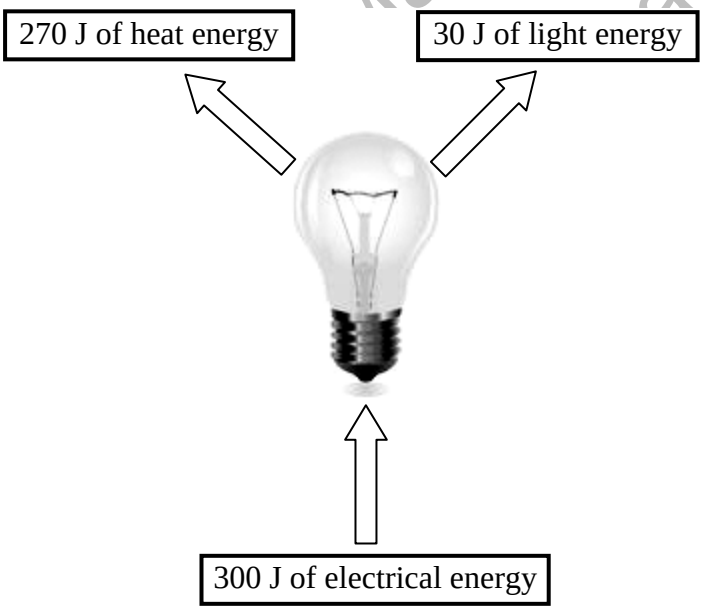
Differentiate between the orbit of a comet and the orbit of a planet in THREE points.

S. No.	Orbit of a Comet	Orbit of a Planet
1		
2		
3		

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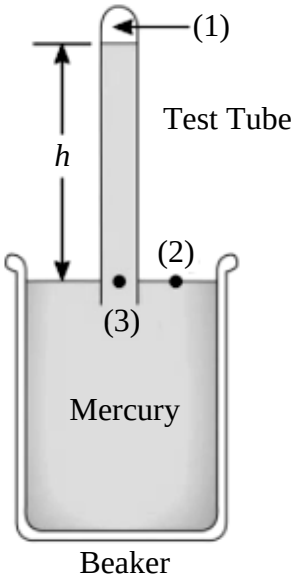
Q.6. (Total 3 Marks)

Calculate the efficiency in percentage of the light bulb shown in the given diagram.



Q.7. (Total 3 Marks)

In the given diagram, a test tube filled with mercury is inverted into a beaker that is also filled with mercury. Assume that the closed end of the test tube is a vacuum.



Compare the pressure at points (1), (2) and (3) for the given diagram.

Q.8. (Total 4 Marks)

If the length of a metallic conductor is 6 m; its area of cross section is 18 m^2; temperature difference between two surfaces of the conductor is 50 K and the rate of heat flow through the conductor is 40×10^3 W, then calculate the thermal conductivity of the conductor.

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Q.9. (Total 5 Marks)

EITHER

a. A body is moving with initial velocity (A) and after a certain time (t), its velocity becomes (B), the distance covered during this time is (S); using the given information derive Newton's equation of motion $2aS = B^2 - A^2$. (5 Marks)

OR

b. Seatbelts and airbags are two safety features provided in cars to protect passengers during any accident.

i. Explain these features with the help of the phenomenon of momentum. (3 Marks)

ii. How can you connect the same phenomenon explained in part (i), when a person jumps from height? (2 Marks)

Q.10. (Total 5 Marks)

EITHER

a. 'Renewable energy sources are better than non-renewable sources.'

Justify in FIVE points the given statement in the context of Pakistan. (5 Marks)

OR

b. Submarines, as compared to ships are designed in such a way that they can either float or submerge under water.

i. Name and state the principle on which submarines work. (2 Marks)

ii. Explain the working of submarines with reference to the principle identified in part (i). (3 Marks)

Q.11. (Total 5 Marks)

EITHER

a. 'Evaporation takes place at all temperatures whereas boiling takes place at a fixed temperature under a given pressure.' Explain the given statement with reference to the concept of evaporation and boiling point.

OR

b. Describe the process of transfer of heat by radiation with reference to infrared radiation and energy of electrons in different orbits.

END OF PAPER

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